

Predicting Risk of Heart Disease (Team 3)

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Abstract—With how high cause heart disease can be towards deaths in the United States, what are key contributors to heart disease risk and heart attacks, and can we accurately determine if someone is prone to these based on given features? We apply and optimize classification machine learning algorithms such as Decision Trees, Random Forest, Logistic Regression, and SVM to achieve this goal.

I. INTRODUCTION

Heart disease is the leading cause of death in the United States of America, according to the Center for Disease Control and Prevention (CDC). There are many factors that contribute to someone having a heart attack, such as high cholesterol, smoking, alcohol, and many more. With this said, it is important to identify early signs of at-risk individuals to enable preventive interventions and reduce the nation's long-term healthcare burden.

This project frames heart disease prediction as a binary classification project: given a set of behavioral, demographic and health features collected from this dataset, is it possible to accurately predict if a respondent has experienced a heart attack? This dataset contains at least 445,000 records and 40 attributes covering self-reported health status, lifestyle behaviors, existing comorbidities, and preventive care utilization. The target variable HadHeartAttack shows a significant class imbalance of 16.6:1, with 416,959 saying No and 25,108 saying Yes, presenting a core methodological challenge.

Our solution pipeline includes thorough exploratory data analysis, missing-value imputation, label encoding of categorical features and evaluation of multiple classifiers. As initial baselines we implement Decision Tree, Random Forest, Logistic Regression and Bayesian Learning. These are some of the algorithms determine if someone is likely to fall to heart disease.

This report is as follows: Section I is the introduction, explanation of dataset and how the results were conducted. Section II surveys related work on ML-based heart disease prediction. Section III describes our dataset, preprocessing, algorithms, and preliminary results for each model. Section IV compares model performance. Section V outlines future directions, and Section VI is the conclusion.

Source: <https://www.kaggle.com/datasets/kamilpytlak/personal-key-indicators-of-heart-disease/>

A. Problem Statement

Given how high cause heart disease can be towards deaths in the United States, what are key contributors to heart disease

risk and heart attacks, and can we accurately determine if someone is prone to these based on given features?

B. Dataset Description

This dataset is composed of 400,000 adults participating in a survey conducted by the CDC regarding their health conditions. The survey information was reduced by the user Kamil Pytlak, on kaggle. It consists of 40 features and comes with two different versions, one with NaN values and one without. For the sake of doing our own work we decided to choose the version with missing values especially since the way the non missing values data was treated was just to drop them all when we could treat those values differently potentially. The data was collected in a 2022 survey from people across the 50 states. The features and topics of the dataset relating to health conditions are related to or indicators of heart disease or risk of heart attacks. The dataset is primarily focused on classification as the target variable is a binary "HadHeartAttack" which can be treated as an at risk or has heart disease. It is also mentioned that the classes are imbalanced so that they need to be considered. The features are varied in terms of datatypes being both qualitative and quantitative.

C. Machine Learning Algorithms

The machine learning algorithms that will be used to perform classification on this dataset are Decision Tree, Random Forests, Logistic Regression, and SVM. We will use randomized search cross validation on the models so that we can conduct tuning on the hyperparameters to make sure that these algorithms have the best performance possible and have efficient computation times with a large dataset.

D. Experimental Results

We were able to discover that our models had relatively high performance and accuracy for a dataset of this size and scale and were able to find the most optimized parameters for determining if someone was at risk of heart disease. We used calculations such as precision, recall, and F1 score to determine these results and later on will compare the results of models between each other. What matters most is our accuracy in general for predictions is higher than randomly guessing a result which would be anything above a 50/50 or 50% accuracy. For decision tree we had a prediction accuracy of roughly 82% and recalls of roughly 77 and 82% for predicting heart risk and no heart risk respectively. For random forest we had a prediction accuracy of roughly 80% and recalls of

roughly 79 and 81% for predicting heart risk and no heart risk respectively.

E. Related Work Comparison

Compared to a lot of attempts at modeling off of this dataset on kaggle, the work regarding methods and techniques do not differ that much as this is a classification problem and ultimately similar algorithms will be used. Where our solutions different potentially compared to some other existing solutions as well as improve upon would be with how we approach hyperparameter tuning as well as further implementation of preprocessing such as filling in missing values instead of dropping them, scaling as well as label encoding. Feature importances and reduction are also taken into account.

II. RELATED WORK

The related work that we discovered for our problem statement was able to be found very quickly due to kaggle's resources. Attached to the dataset's page itself on the kaggle website, there is a section of notebooks that people have shared and worked on that do cover prediction for heart attacks and heart disease. For the most part, very popular methods are logistic regression, decision trees and random forests classifiers which is what we also are using. Provided below is a list of works related to this dataset for finding risk of heart disease.

Source: <https://www.kaggle.com/datasets/kamilpytlak/personal-key-indicators-of-heart-disease/code>

III. OUR SOLUTION

This section elaborates on our solution to the problem.

A. Description of Dataset

This dataset is composed of 400,000 adults participating in a survey conducted by the CDC regarding their health conditions. The survey information was reduced by the user Kamil Pytlak, on kaggle into a dataset. The dataset is primarily focused on classification as the target variable is a binary "HadHeartAttack" which can be treated as an at risk or has heart disease. The features are varied in terms of datatypes being both qualitative and quantitative.

In terms of preprocessing and how we need to handle the data, a few things needed to be taken into account with feature engineering. The biggest, most noticeable problem when choosing our dataset was that it has missing values. Technically the creator of this dataset has two different versions, where one does not have missing values but the method in which it was dealt with was just dropping them. What we decided to do was fill in the missing data. For numerical features, we would fill in any NaN values with the median, as this would allow us to not have to worry about outliers skewing the average which can happen with mean. For categorical features, we decided to give it a placeholder "Unknown" value. We also decided to drop any columns that have too many missing values by doing a threshold check of 40 percent. Essentially, if a feature has 40 percent or more of the rows comprised of missing values, we can drop them.

Another problem we needed to take into account was how categorical values are string values, which need to be converted

to a numerical one for use in our models. This was dealt with through the use of label encoding, where each unique category value in a feature was given a unique integer value. We also need to normalize the numerical features in the dataset which can be accomplished using the StandardScaler function in the scikit-learn library.

Another issue we need to look into potentially is multi-collinearity, which is occurs when features in a dataset are highly correlated, which is a problem for regression models such as our logistic regression one as well as SVM. If one feature is able to be predicted from another, we would have a problem. To solve this we want to create a correlation matrix which checks all of the feature combinations to see what level of correlation we have for them. We are looking for our features to have correlation leaning towards zero and luckily our features do. Because of this, we actually do not need to manipulate our dataset or drop any features. Shown below is a figure of our correlation matrix:

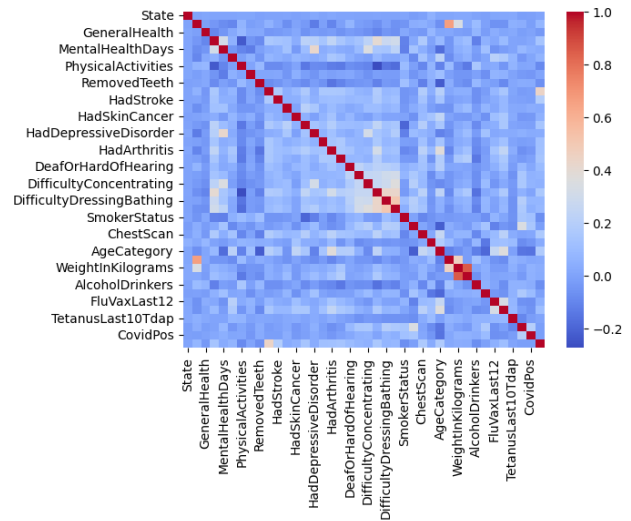


Fig. 1. Correlation Matrix for Our Dataset

It is also mentioned that the classes are imbalanced, there are a significantly more people in this dataset who have not had heart attacks compared to people who do. There are 416,959 people in the dataset who had not had a heart attack (no risk of heart disease) compared to the 25,108 who had any, which is roughly 6% of the dataset. The main way we deal with this is through downsampling which is a method in which we cut down on the majority class representation on our dataset so that we can have a more balanced ratio between the positive and negative predictions. We decided to choose a 1:1 ratio between no heart risk and heart risk. Here is a code example:

```
df_downsample = resample(
    df_majority ,
    replace=False ,
    n_samples=len(df_minority) ,
    random_state=42
)
```

There are a few other ways this can be dealt with but the way we choose to deal with this later is with the models itself such

as decision tree and random forest for example which have a way to weight classes if there is imbalance, making it so that the less represented side in the dataset can get more focus and better accuracy. We also can use accuracy measures such as area under the curve, recall, precision, and F1 score to get a better representation of data regardless of imbalance. While SMOTE could be used, it does not seem like it is necessary and the dataset itself is already very large, to balance the two classes we would need to make essentially 400,00 heart risk fake people, downsampling would be better instead.

To be able to view the dataset and download it, the link will be provided below.

Source: <https://www.kaggle.com/datasets/kamilpytlak/personal-key-indicators-of-heart-disease/>

B. Machine Learning Algorithms

The machine learning algorithms we plan to use for this problem are decision trees, random forests, logistic regression, and SVM. We plan to use all of these because our problem is a classification one where we need to determine if a person is at risk of heart disease or not (heart attack). Individually, we wanted to use decision trees because of the potential of having a readable and visible way to see how predictions are made, as well as the fast performance it can have with the right tuning as well as being able to see feature importance with pruning. Random forests we also chose to use as a comparison for decision tree and also as a way to reduce overfitting potentially compared to just a single tree as well as for feature importance as well.

The main design for decision tree and random forests was very similar in terms of them using the same parameters and process for fitting and modeling. A randomized search cross validation was performed on both of these to find the best choices for hyperparameters such as the max depth of the trees, max leaf nodes, minimum samples split, minimum samples leaf, and the criterion for either gini or entropy. The reason this was decided right away was because we would be able to automate and efficiently choose the most optimized combination of parameters to get the best performance out of them. Other than that, for the estimators for random forest we chose 50, and made sure to have the class weight be balanced so that the dataset does not get negatively impacted by the imbalance between people with heart risk and ones who do not.

As for logistic regression, it is a natural baseline for binary cardiovascular risk classification, as it mirrors the structure of established clinical risk scores. Its linear decision boundary is highly interpretable via coefficient magnitudes and odds scores, allowing clinicians to understand which features increase risk. This algorithm is appropriate for this situation because many cardiovascular risk factors are known to have monotone effects on log-odds of heart disease.

C. Implementation Details

The first implementations we will cover is the decision tree method, we use this because it is a very efficient and commonly used classification model that allows us to have a readable visual for prediction as well as being able to find

important heart disease risk factors. The way we went about this algorithm was to first go through hyperparameter tuning to optimize the decision tree's performance. We used randomized search cross validation to find the best combination of parameters, it runs the model multiple times with these different combinations and based on the highest score chooses one. We also applied stratified k fold into this for the cv as a way to make sure we are using balanced class choices when a part of our dataset is randomly selected. Tuning our decision tree also helps us prune it, by reducing redundant features and reducing overfitting since the tree can get too complex. We also selected the tree to have balanced weights. Doing this instead of fitting and testing the model over and over manually, automates this process and increases computational efficiency with such a large dataset. The following parameter combinations were considered and were fitted and the best results were shown also below the code:

```
params = {
    'max_depth': [5, 10, 15, 25, None],
    'max_leaf_nodes': [20, 50, 100, 200],
    'min_samples_split': [2, 10, 50, 100],
    'min_samples_leaf': [1, 5, 10, 20],
    'criterion': ['gini', 'entropy']
}

cv = StratifiedKFold(n_splits=3, shuffle=True,
                    random_state=42)

random_search = RandomizedSearchCV(
    tree,
    param_distributions=params,
    n_iter=75,
    cv=cv,
    scoring='average_precision',
    random_state=42,
    n_jobs=-1
)
```

Now that we had the best parameter choices ('min samples split': 50, 'min samples leaf': 20, 'max leaf nodes': 100, 'max depth': 15, 'criterion': 'gini'), we could fit our decision tree with these and also set the model to have balanced class weights as our dataset was imbalanced. After we fit the tree, we have a visual representation shown in figure 2:

After all of this, we get our model results of a prediction accuracy of roughly 79 percent meaning that this percentage is the amount of predictions the model got correct. We also had an area under the curve of 87.33 percent which means this percentage is the how well the model can separate between the classes with discrimination. We also had recall, precision, and F1 scores of 79 percent across the board for heart risk and no heart risk. Meaning 79 percent of patients who predicted as actually having heart risk actually were, 79 percent were correctly identified, and our model is well balanced for false positives and negatives.

For feature importance for our decision tree, we notice that HadAngina is the significantly highest feature. When researching this feature, this makes sense, it is related to having chest discomfort due to a lack of oxygen and blood flow to the heart which can lead to heart disease or potentially already a

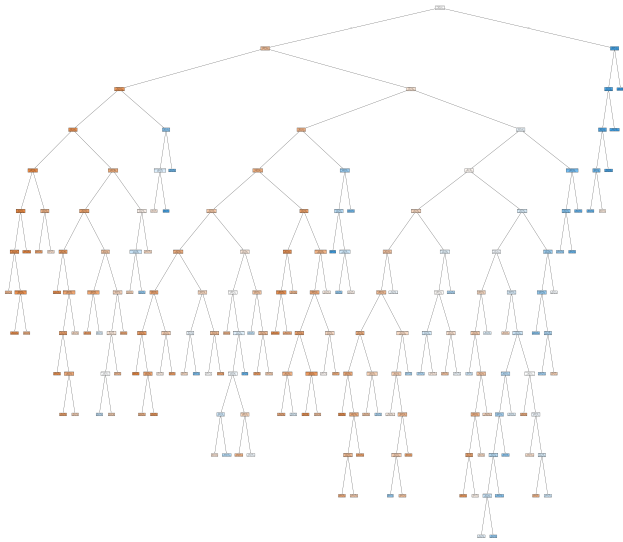


Fig. 2. Pruned Tree

sign of it existing. HadStroke and ChestScan also relate to this. Overall a lot of these features have to do with underlying health conditions, diet, physical activity and physical qualities. Some features such as age, stroke, chestscan and difficulty walking rank among the highest predictors of heart disease. We can conclude from this that the top factors relate to overall health and fitness, which can cause people to self reflect on their day-to-day life. We did notice though that a significant factor that people would think would matter in a model such as alcohol consumption, BMI and smoking don't have as much of an impact. Figure 3 shows the top 15 feature importances:

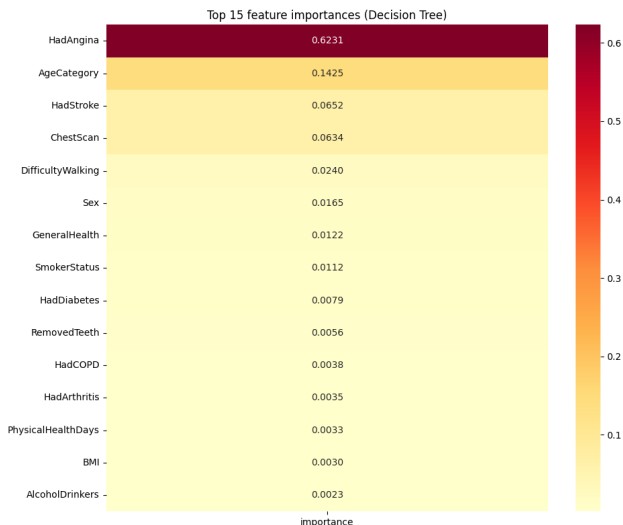


Fig. 3. Top 15 Feature Importances from Decision Tree

For the next model we move to random forest. We go through the a similar process for random forests as we did decision tree. We decided to also use this model because it can handle overfitting by building many trees and take samples from them and make an accurate prediction by taking the average or majority of the best prediction. For handling this

model, we can fit this relatively fast by using the same parameter combination that decision tree had, including balanced weights, we also add 100 estimators. After all of this, we get our results of a prediction accuracy of roughly 80 percent meaning that this percentage is the amount of predictions the model got correct. We also had an area under the curve of 88.43 percent which means this percentage is the how well the model can separate between the classes with discrimination. We also had recall, precision, and F1 scores of 80 percent across the board for heart risk and no heart risk. Meaning 79 percent of patients who predicted as actually having heart risk actually were, 79 percent were correctly identified, and our model is well balanced for false positives and negatives. This compares very similarly to decision tree which makes sense as the parameters were tuned the same way which already helped with overfitting in the previous scenario. There are slight improvements though however. Here is the code for our random forest model below:

```
forest = RandomForestClassifier(
    n_estimators = 100,
    min_samples_split=50,
    min_samples_leaf = 20,
    max_leaf_nodes = 100, max_depth = 15,
    criterion = 'gini',
    class_weight="balanced",
    random_state=42)
forest.fit(X_train, y_train)
```

For feature importance related to random forest, by looking at the top 15 features that impacted our random forest model, we can once again highlight what factors people should look out for regarding heart diseases. HadAngina is again, the significantly highest feature. HadStroke and ChestScan also are important features. Overall a lot of these features have to do with underlying health conditions, diet, physical activity and physical qualities. Some features such as age, stroke, diabetes and removed teeth rank among the highest predictors of heart disease. We can conclude from this that the top factors relate to overall health and fitness, which can cause people to self reflect on their day-to-day life. We can also see other top features like difficulty with physical activities and chest scans are ranked towards the top, strengthening the argument that physical health indicators are strong predictors of heart disease risk.

There are also some outliers in our analysis including groups like SmokerStatus, which we usually associate with heart disease pretty heavily in our society, and according to the heatmap, it is ranked near the bottom. One more interesting finding includes the relevance of previous diseases, we see numerous features in the heatmap towards the top to middle that revolve around experiencing a different type of health issue. BMI not being a contributing factor or even alcohol consumption is also highly interesting in this scenario. Figure 4 provides the top 15 feature importances for random forest:

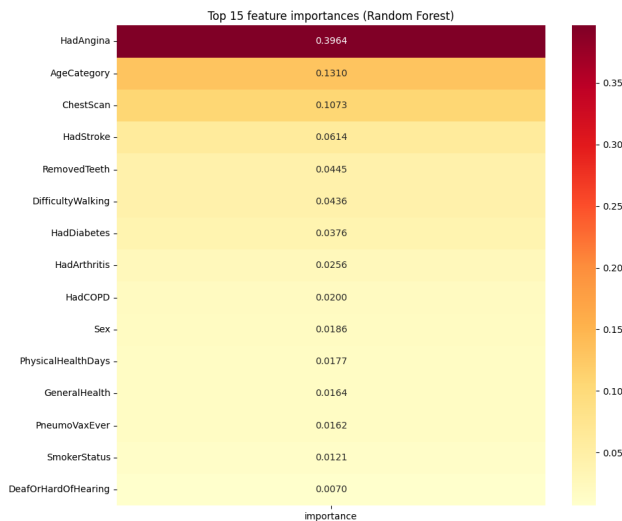


Fig. 4. Top 15 Feature Importances from Random Forest

After performing these calculations, we discovered the following: for decision tree, we had a prediction accuracy of roughly 82% and recalls of roughly 77 and 82% for predicting heart risk and no heart risk, respectively. For random forest we had a prediction accuracy of roughly 80% and recalls of roughly 79 and 81% for predicting heart risk and no heart risk respectively.

The next implementation we will cover for the dataset will be logistic regression. This model is wrapped in a scikit-learn Pipeline (StandardScaler to Logistic Regression) to prevent data leakage - the scaler is fitted only on training data and applied to test data. Key parameters to train the model will look like this:

```
pipe = Pipeline([
    ('scaler', StandardScaler()),
    ('clf', LogisticRegression(
        class_weight='balanced',
        max_iter=1000,
        solver='lbfgs',
        C=1.0,
        random_state=42,
        n_jobs=-1
    ))
])

pipe.fit(X_train, y_train)
```

This code will be applied on the downsampled 1:1 balanced training set which includes roughly 40,000 samples. Five-fold stratified cross-validation confirms very stable performance: fold AUC scores of 0.8814, 0.8846, 0.8804, 0.8913, 0.8900, yielding CV AUC = 0.8855 ± 0.0044 . The code snippet of 'class_weight='balanced' combined with the 1:1 downsampled set, ensures the model does not trivially predict the majority class. With the balanced test set (50% positive), precision and recall are directly comparable across classes - achieving precision 0.78/0.82 and recall 0.84/0.76 for No HA/Heart-Attack respectively.

These findings offer us our odds-ratio insights. HadAngina

dominates with OR = 3.249 - by far the strongest single predictor, reflecting that chest pain from reduced coronary blood flow is a direct precursor to heart attack. AgeCategory (OR=1.855) is the second strongest, consistent with the age-dependency of cardiovascular disease. HadStroke(OR=1.468) and Sex(OR=1.441) follow, with SmokerStatus (OR=0.816) showing a negative coefficient - likely reflecting survivor bias or complex interactions effects in this cross-sectional survey.

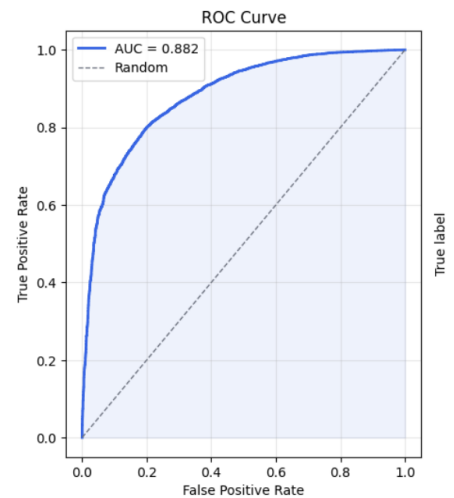


Fig. 5. ROC curve with a AUC of 0.882

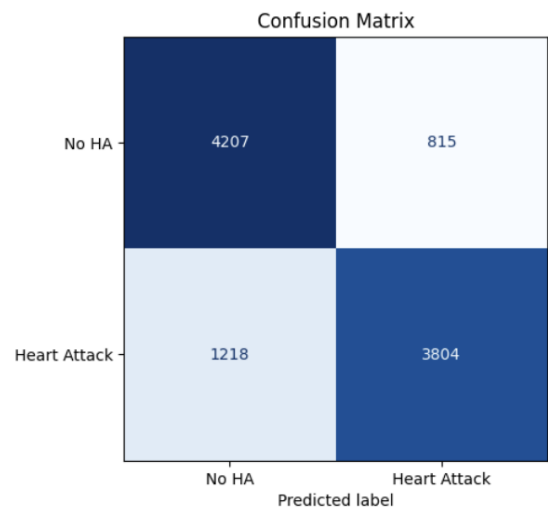


Fig. 6. Confusion Matrix for Logistic Regression, showcasing true positive/true negative counts

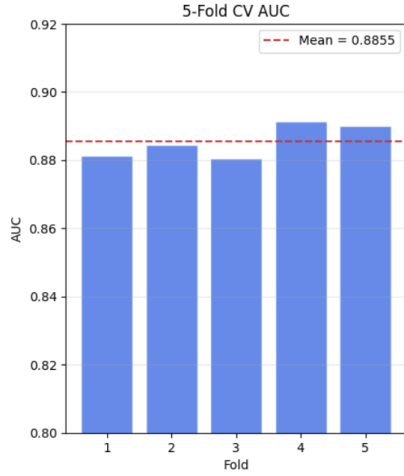


Fig. 7. 5-fold CV AUC bar chart where the mean is 0.886 and standard deviation is 0.004

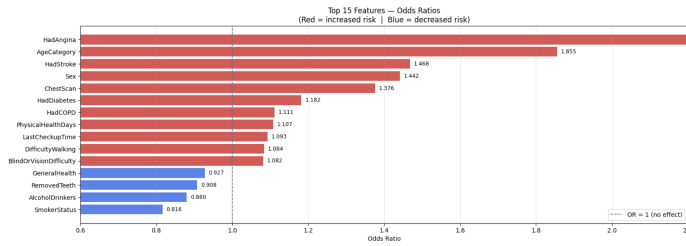


Fig. 8. Top 15 Features with Odds Ratios, where red indicates increased risk and blue indicates lowered risk

Lastly, the last model we implemented for our analysis is our Support Vector Machine model. SVMs are supervised learning models that find an optimal decision surface (hyper-plane) in feature space to classify instances. The best approach for our large dataset was to utilize the LinearSVC estimator over the standard SVC. When attempting to use standard SVC, the runtime of the SVM algorithm was almost unusable as it would take up to 20 minutes to generate results. By implementing LinearSVC, we saw more efficient results with a reasonable runtime, solely based on LinearSVC using a linear kernel to optimize the dataset.

```
svm = LinearSVC(
    C=1.0,
    random_state=42,
    max_iter=2000,
    class_weight='balanced')
```

This implemented model was trained using 80% of the dataset with the leftover 20% as the testing set. Before training, we ensured that all numerical features were standardized using StandardScaler, and categorical features were encoded using LabelEncoder. Pre-processing is incredibly important when setting up our SVM, feature scaling needs to be scaled properly to produce accurate results. For example, our BMI values typically ranged from 10 to 60, while MentalHealthDays ranged from 0 to 30, and HeightInMeters ranged from 1.0 to 2.5. As you can see, the scaling needed to be standardized to

prevent features with larger numeric ranges from dominating the model.

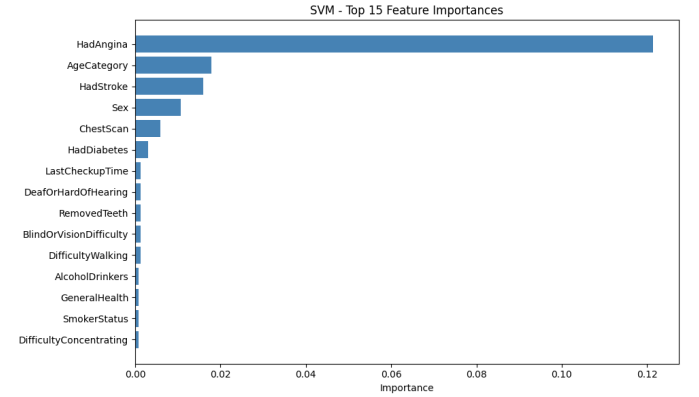


Fig. 9. Top 15 Feature Importances from SVM

When dissecting the feature importance chart, it revealed a lot of key information and predictors that strongly influenced the models predictions. As seen in the figure, HadAngina can be characterized as they key feature, with a much larger importance score than the other features. According to background research, our top features align with medical expectations as Angina is considered one of the most prevalent factors in heart attacks. Other highly known features like AgeCategory, HadStroke, and Sex, were included at the tops of the features chart also aligning with medical expectations. Features like HadDiabetes and ChestScan represent aspects of a patients prior medical history, emphasizing the importance of past health conditions and utilizing predictive performance in medicine.

IV. COMPARISON

The logistic regression model's AUC of 0.8847 sets a strong baseline that ensemble methods must surpass. Its 76% minority-class recall is promising for a screening tool as three quarters of all true heart attack cases are flagged for further evaluation. The moderate precision (0.82) is an acceptable result in this context, as downstream clinical testing would least-likely filter false positives. A naive majority-class classifier would achieve 94% accuracy but 0% heart-attack recall and AUC = 0.50, illustrating why accuracy alone is a misleading metric under severe imbalance. Odds-ratio insights from logistic regression will be cross-validated from other algorithms to feature importances to assess consistency of top predictors such as AgeCategory, HadAngina, Sex and HadStroke.

The decision tree model's AUC also sets a strong baseline of 0.8733 and has a high minority-class recall too of roughly 79%. The precision is not too far off from the logistic regression being 79%. Random forest has a similar AUC for both of these in which it is 0.8843. The recall is also promising for both of these being 80% and has a high precision too of roughly 80 too. Both of these models show high feature importances sharing HadAngina, AgeCategory, ChestScan and HadStroke.

The support vector machines' model outputted a AUC score of 0.8817 along with a 80% overall accuracy. All 3

models had very similar precision outputs, existing somewhere in the range of .77-.83 across both classes. The SVM had a heart attack precision of .83 and a no-risk precision of .77, with corresponding F1 scores around .79 and .81. The heart attack recall is slightly lower than both decision tree and random forest, making the SVM less sensitive to detecting true heart attack cases. However, the average of precision, recall and F1 score shows that the SVM performs consistently across both classes.

V. FUTURE DIRECTIONS

If we were given 6-12 months, we would pursue a few different ways to handle and further optimize our model performance. A major problem we faced was the data imbalance between people in this dataset who were at risk of heart disease, and those who weren't. We did handle this through downsampling as well as in our models itself with parameter tuning, but other methods potentially such as SMOTE could have improved performance further. Other models potentially being used could also be considered in the future such as XGBoost. Feature reduction should be looked into more potentially considering that Angina was a huge factor in all of our models, potentially removing this and letting other features have more room to grow could allow higher performance.

VI. CONCLUSION

This report presents the implementation and evaluation of for machine learning classifiers for heart disease prediction on the 2022 dataset. All four models achieve AUC-ROC in the narrow range of 0.873-0.884 and accuracy of 79-80% on a balanced 1:1 test set, demonstrating that the survey features contain strong and consistent predictive signal for heart attack risk. Across all models, we also noticed that "hadAngina" was the significantly most prominent feature that determined whether a person was at risk of heart disease or not. Ultimately, we can use our models and findings as a relatively accurate way to signal and determine if a person is at risk of heart disease and what qualities they have that they can potentially improve upon to prevent this from happening.

REFERENCES

Dataset: <https://www.kaggle.com/datasets/kamilpytlak/personal-key-indicators-of-heart-disease/>

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